Total No. of Printed Pages: 3

SUBJECT CODE NO: - RNP-8007
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group B]Comp., AI &ML,EEE
Examination January/February 2025
Basics of Electrical Engineering

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Question No. 2, 3, 4, 5, 6 and 7.

SECTION A

Q1 Solve any 10 from the following questions. **20**

1. The total opposition offered to the flow of alternating current by a circuit is called
   A. Reactance B. Impedance
   C. Resistance D. Total resistance

2. The ratio of r.m.s. value to the average value of an alternating quantity is known as Ampere
   A. Form Factor B. Peak Factor
   C. Maximum Factor D. Current

3. Line voltage in star connected three phase...........times phase voltage.
   A. 3 time B.√3
   C. Equal D. 2 Times

4. Line current in delta connected three phase ....times line current.
   A. 3 time B.√3
   C. Equal D. 2 Times

5. The value of form factor of sinusoidal waveform is
   A. 1.11 B. 1.21
   C. 0.70 D. 0.63

6. The unit of electrical field Magneto motive force will be
   A. AT B. Web
   C. Ampere D. volt

7. The conductor and flux having angle between them 90 then flux link
   A. Maximum B. Minimum
   C. Equal D. Zero

8. The ratio of maximum value to the r.m.s. value of an alternating quantity is known......................
   A. Zero B. Peak Factor
   C. Form Factor D. None

9. The flux variations being brought about by variations
   A. Density B. voltage
   C. current D. Power

10. An ideal voltage source is that which has an internal resistance
    A. Zero B. Low
    C. High D. Infinite

11. The energy used in one hour at the rate of 1kW is known as......
    A. 1 Unit B. 1 watt
    C. 1 Newton D. 1 volt

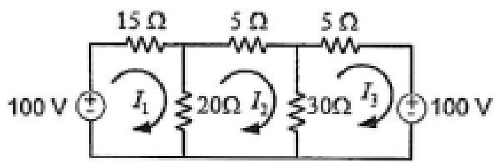
12. In Kirchhoff's first law Ei = 0 at the junction is based on the conservation of
    A. Charge B. Current
    C. Energy D. Voltage

13. The resistance of a material is most commonly determined by four factors- length, cross-sectional area, type of material and ....
    A. Current B. Voltage
    C. Power D. Temperature

SECTION B

Q2 a) Prove the Series and parallel combination of resistor with circuit diagram. **05**
b) Expression for value of α at different temperature in detail. **05**

Q3 a) Find loop Current I1, I2 & I3 by Mesh Analysis. **05**



b) Explain Superposition theorem with an example. **05**

Q4 a) Comparison between Magnetic and Electric Circuits. **05**
b) A solenoid have 800 turns supply with 30 volt of resistance is 6 ohm, length of solenoid is 16 cm with area is 6 sq.cm, relative permeability is 800 find 1) MMF 2) Magnetic field intensity 3) Flux density 4) Reluctance **05**

Q5 a) Prove the Coefficient of Coupling with equation. **05**
b) Prove the Capacitors in Series with detail. **05**

Q6 a) Explain R-L-C Series A.C. Circuit with phasor diagram **05**
b) Explain R-C Series A.C. Circuit with phasor diagram **05**

Q7 a) Explain & Draw the phasor diagram for Star connected phase voltage and line voltage **05**
b) Explain & Draw the phasor diagram for Delta connected phase current and line current **05**